AFF–1049

Sub. Code
4MBO1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

PLANT DIVERSITY – I
(ALGAE, FUNGI, LICHENS AND BRYOPHYTES)

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Para amylum.

2. Cyanobacteria.

3. Kelp.

4. Anisogamy.

5. Asci.

6. Pycridium.

7. Fungal pheramone.

8. Soredia.

9. Columella.

10. Naidiata Lanceolata.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a concise account on algae as pollution indicator.

Or

(b) Write a comparative account on cyanophyceae with Rhodophyceae.

12. (a) Write the salient features of marine algae.

Or

(b) Write a brief account on algae as a source for single cell protein.

13. (a) Why are most fungi confined to a specific ecological niche?

Or

(b) Write notes on edible fungi.

14. (a) State the significance of pheramones in the reproduction of fungi. Name at least three fungi which produce them.

Or

(b) Compare the para sexual cycle with sexual cycle by giving examples.

15. (a) The foot of the sporophyte of bryophytes remain embadded in the tissues of gametophyte why? Give reasons.

Or

(b) Write an account on fossil Bryophytes.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write in detail on the ultrastructure of prokaryotic and Eukaryotic algal cell components.

17. Explain the phenomenon 'Alternation of Generation' in algae.

18. Why majority of the Basidiomycetes exists in the wood? Discuss the mating system in Basidiomycetes.

19. Describe the various reproductive structures associated with the thalli of Lichens.

20. Bryophytes are ecologically very important. Yet significant, these play a silent but quite a vital role in the nature. Discuss.

AFF–1050

Sub. Code
4MBO1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

PLANT DIVERSITY – II

(PTERIDOPHYTES, GYMNOSPERM AND PALEOBOTANY)

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Mycorrhizal fungi.
2. Gemmae.
3. Plectostele.
4. Leptosporangiate pteridophyte.
5. Pycnoxylic wood.
6. Haplocheilic stomata.
7. Ginkgo biloba.
8. Ephedrin.
9. Dendrochronology.
10. Casts.

Part B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

11. (a) How will you classify the pteridophytes by following G.M. Smith (1971)?

Or

(b) Explain the organization of spore producing organs of Equisetales.

12. (a) Give an account of seed habit in Pteridophytes.

Or

(b) Describe the structure of any two types of steles met within pteridophytes.

13. (a) Analyse the general characters of Gymnosperms.

Or

(b) Explain the external and internal structure of Pteridospermales.

14. (a) Discuss the organization of male strobili of Cycadales.

Or

(b) List down any five economic importance of Gymnosperm.

15. (a) Analyse the structure of the cupule of *Lyginopteris*.

Or

(b) Examine the facets of paleopalynology.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Critically evaluate the reproduction of Isoetales.

17. Write an essay on the soral evolution of Pteridophytes.

18. Compare the external structure of the members of Cordaitales.

19. Discuss the organization of strobili in Gnetales.

20. Describe the process of fossil formation.

AFF–1051

Sub. Code
4MBO1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

MICROBIOLOGY, PLANT PATHOLOGY AND IMMUNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short notes on the followings :

1. Virions.
2. Differentiate between prokaryotes and eukaryotes.
3. Budding.
4. Batch culture.
5. Gram positive bacteria.
6. Epidemic disease.
7. Biochemical defense mechanism.
8. Rous sarcoma virus.
9. IgG.
10. Causal organism and control measures of tikka disease.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain five kingdom system classification by whittaker.

Or

(b) Discuss the replication and transmission of virus.

12. (a) Mention the general characteristics of cyanobacteria.

Or

(b) With a neat diagram describe the ultrastructure of bacteria.

13. (a) What are the different methods of studying plant diseases?

Or

(b) Briefly write the characteristics of mycotoxins.

14. (a) Elucidate the causal organisms and symptoms of rice blight.

Or

(b) Explain the causal organism and symptoms of Tobacco mosaic disease.

15. (a) Discuss the antigen and antibody interaction.

Or

(b) Briefly mention the principle and methodology of immunoelectrophoresis.

Part C (3 × 10 = 30)

Answer any **three** questions only.

16. Distinguish the general characteristics between prokaryote and eukaryote.

17. Summarise the economic importance of bacteria.

18. Discuss the host parasite interactions.

19. Illustrate the causal organism, symptoms and control measures of citrus canker disease.

20. Write a general, account on immune system.

AFF–1052

Sub. Code
4MBO1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

GENETICS, PLANT BREEDING AND BIOSTATISTICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short notes on the following.

1. Pseudoallele.
2. Law of segregation.
3. Transposans.
4. Bacteriophage.
5. F_1 Particle.
6. Hardy – Weinberg law.
7. Autopolyploidy.
8. Asexual propagation.
9. Arithmetic mean.
10. Mode.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss the interaction of genes with suitable example.

Or

(b) Explain the non-disjunction of sex chromosome.

12. (a) Elucidate the modern concept of genes.

Or

(b) How would you detect the mutation in bacteriophage?

13. (a) What is male sterility? Explain it with suitable example.

Or

(b) Write short note on gene frequencies.

14. (a) List the merits and demerits of self-fertilized plant breeding.

Or

(b) Briefly give an account on plant breeding work in India with special reference to rice.

15. (a) Mention the general rules for the construction of table.

Or

(b) Explain the procedure for χ^2 test. What are the applications of chi-square test?

Part C (3 × 10 = 30)

Answer any **three** questions only.

16. Illustrate the different theories of sex determination. How sex determination occur in plants?

17. Define Mutagen. Discuss the various physical and chemical mutagens and their mode of action.

18. Give an overview of genome of mitochondria.

19. Describe the role of polyploidy in plant improvement.

20. Elucidate the properties of regression line. Differentiate between correlation and regression.

AFF–1053

Sub. Code
4MBO1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

***Elective* — MUSHROOM CULTIVATION AND VERMICULTURE**

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short note on the following.

1. Stipe.
2. Tray method in mushroom cultivation.
3. Economics of mushroom cultivation.
4. Compost.
5. Pot method of vermi composting.
6. Reclamation by earthworms.
7. Vermi filters.
8. Role of earthworm in soil fertility.
9. Spawn.
10. Important South Indian species of earthworm.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What are medicinal mushrooms? Explain.

Or

(b) Give an overview of the present status of mushroom industries in India.

12. (a) Elucidate the technique of spawn production.

Or

(b) How will you control pests and nematodes in mushroom cultivation?

13. (a) Discuss the role of earthworms in soil fertility.

Or

(b) Explain the food habits and excretion of earthworms.

14. (a) What is worm cast? Mention its importance.

Or

(b) What are the problems faced in vermi compost preparation?

15. (a) Explain the method of vermi wash production and also mention its uses.

Or

(b) How vermi compost are useful for crop production?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Give an overview of cultivable edible mushrooms.

17. Elaborate the technique of cultivation of white button mushroom.

18. Describe the life cycle of earthworm.

19. Outline the collection and maintenance of earthworms for vermi composting.

20. Describe the large scale manufacture, packaging of vermi compost and marketing.

AFF–1054

Sub. Code
4MBO1E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

Elective – ORGANIC AGRICULTURE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short note on the followings :

1. Organic farming
2. Shelter zones
3. Soil fertility
4. Biofertilizers
5. Farming systems
6. Soil productivity
7. Weeds
8. Biopesticides
9. Organic certification
10. Labelling.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short notes on land management.

Or

(b) Mention the future prospects of organic farming.

12. (a) Explain briefly about nutrient recycling.

Or

(b) Write the technique for compost preparation.

13. (a) Explain about crop rotation.

Or

(b) Discuss multiple and relay cropping systems.

14. (a) How to control the weeds?

Or

(b) Discuss the insect pest management.

15. (a) Explain the certification process.

Or

(b) Discuss certification in India and around the world.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Define agro forestry. Explain the different kinds of agro-forestry.

17. Illustrate the techniques of vermicompost preparation. Mention the advantages of earthworms.

18. Discuss intercropping in relation to maintenance of soil productivity.

19. Give a detailed notes on biological agent in agro farming.

20. Explain socio-economic aspects of local and national organic farming markets.

AFF–1055

Sub. Code
4MBO2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

TAXONOMY OF ANGIOSPERMS AND ETHNOBOTANY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Dioscoriodes.
2. Linnaean system of classification.
3. Cladistics.
4. National herbaria.
5. Dichotomous key.
6. Tautonym.
7. Aggregate fruit.
8. Apocarpous.
9. Ethnomedicine.
10. Alkaloid.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Mention the merits and demerits of natural system of classification.

Or

(b) Trace out the origin of angiosperms.

12. (a) What is biosystematics? Note on its steps.

Or

(b) Explain the role of BSI in botanical research.

13. (a) Write an account on botanical keys and their types.

Or

(b) Discuss the effective and valid publication based on ICBN.

14. (a) Describe the vegetative and floral features of capparidaceae with suitable diagrams.

Or

(b) Sketch out and explain the inflorescence of Poaceae.

15. (a) Explain about tribal medicine and its importance.

Or

(b) Prove that local knowledge system on conservation of medicinal plants.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Compare and contrast the Bentham and Hooker and Engler and Prantl systems of classifications.

17. Justify the chemotaxonomy as good system for resolving taxonomic problems with evidence.

18. List out the principles of ICBN and types of typification.

19. Give a comparative account on the families Rubiaceae and Boraginaceae.

20. Explain the principles, diagnostic methods and medicinal preparation of Siddha system.

AFF–1056

Sub. Code
4MBO2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

ANATOMY, EMBRYOLOGY AND MORPHOGENESIS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Quiescent centre.
2. Lignin.
3. Periderm.
4. Leaf gap.
5. Sporogenous tissue.
6. Ubish bodies.
7. Apospory.
8. Parthinogenesis.
9. Morphogenesis.
10. Differentiation.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Comment on theories of shoot apex organization.

Or

(b) Describe the structure of phloem.

12. (a) Draw and explain the primary structure of dicot root.

Or

(b) Write a note on the secondary growth of dicot stem.

13. (a) Briefly explain the development and structure of microsporangium.

Or

(b) Write a note on the types of ovule present in angiosperms.

14. (a) Describe the development of monocot embryo.

Or

(b) Write short account on polyembryony.

15. (a) Elaborate the morphogenetic factors.

Or

(b) Write about the plant galls and their importance in morphogenesis.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the types, structure and functions of vascular cambium.

17. With appropriate diagram explain the root stem transition.

18. What do you understand by monosporic, bisporic and tetrasporic embryosac? Describe the development of polygonum type of embryo sac.

19. Describe the various types of endosperms found in angiosperms.

20. Explain the molecular basis of morphogenesis.

AFF–1057

Sub. Code
4MBO2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

MOLECULAR BIOLOGY, EVOLUTION AND CELL SIGNALING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Z-form of DNA.
2. Genetic code.
3. Exons.
4. Heterochromatin.
5. Photoreactivation.
6. Sympatric speciation.
7. Parismony.
8. AMP.
9. Elicitors.
10. Integrins.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the different types of DNA replication.

Or

(b) Mention about RNA processing in Eukaryotes.

12. (a) What are the role of chromatin in gene expression?

Or

(b) Write about genetic recombination.

13. (a) Explain the concepts of molecular evolution.

Or

(b) Note on gene dublication and divergence.

14. (a) Give an account on regulation of signaling pathways.

Or

(b) Mention about light signaling in plants.

15. (a) List out the general principles of cellular communication.

Or

(b) Write about neurotransmission and its regulation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the structure and functions of different types of RNA.

17. Write an account on mechanisms of DNA repair.

18. Describe the molecular tools in phylogeny, classification and identification.

19. Give an account on bacterial chemotaxis and quoram sensing.

20. Write in detail about intercellular and intracellular protein sorting with suitable examples.

AFF–1059

Sub. Code
4MBO2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

Elective – ALGAL TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Hold fast

2. Chlorellin

3. Bio-pond

4. Dinoflagellates

5. *Fucus*

6. Gametophyte

7. Alginates

8. Net culture

9. Seaweed cosmetics

10. Biopolymer.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the taxonomic identification methods of fresh water algae.

Or

(b) Give an account on chemical constituents of culture media for marine algae.

12. (a) Justify the photobioreactor technology for micro-algae cultivation.

Or

(b) Explain the source of contaminants in algal cultivation.

13. (a) Classify the seaweeds based on morphology.

Or

(b) Mention the influence of growth regulators on morphogenesis.

14. (a) List out the objectives of marine algal cultivation.

Or

(b) Mention about marketing of seaweeds.

15. (a) Give an account on seaweed fertilizer.

Or

(b) Justify the seaweeds as industrial raw materials.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write an account on laboratory culture method of micro-algae.

17. Explain the harvesting and oil extraction method from seaweeds.

18. Give an account on cell wall polysaccharides is seaweeds and their significance.

19. Elaborately explain the any two marine algal cultivation methods studied by you.

20. Write an essay on seaweeds are sources of useful products.

AFF–1060

Sub. Code
4MBO2E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

Elective – COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Nucleic acid database tools
2. Protein Lounge.
3. High throughput sequencing.
4. Non-Redundant Protein Database
5. Uniport database.
6. Evidence Based Health Practice.
7. Saline level sensor.
8. Artificial intelligence.
9. Boolean nodes.
10. Ordered values.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Examine the various types of data.

Or

(b) Give a concise note on the composite protein database.

12. (a) Briefly explain the shotgun metagenomics principle.

Or

(b) Explains how SEED is applied in the Environmental Genomics.

13. (a) Evaluate the RASMOL tool and its specific application.

Or

(b) List down the varying utility features of PUB-MED.

14. (a) Explain the advantages of maintaining Electronic Health Records.

Or

(b) Discuss the Ethical issues to be adapted in the field.

15. (a) Write a brief account on the Principles and applications of Computational Biology.

Or

(b) Bring out the varying hierarchies of softwares.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Give a detailed account on the Specialized Genome databases.

17. Explain the concept and method of Metagnomics and its applications.

18. Comment on 'Gene sequencing tools' and their unique features.

19. Explain the management procedures adapted in Biomedical data and their application.

20. Write an essay on the Biological Data Transformation System into digital forms.

AFF–1061

Sub. Code
4MBO2E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

***Elective* — FOOD PROCESSING TECHNOLOGY**

(CBCS – 2014 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define the following :

1. Body Building Foods.

2. BMR.

3. Stabilizers

4. Synthetic sweeteners

5. Subcutaneous.

6. Soft diet.

7. Curing.

8. Sealers.

9. HACCP.

10. FSSA.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the recommended dietary allowances of Indians.

Or

(b) Give an account of the nutrition in pregnancy.

12. (a) Explain any two food additives.

Or

(b) Describe the role of dietary fibres.

13. (a) Explain the factors modifying the drug effects.

Or

(b) Describe the importance of diet therapy.

14. (a) Explain the membrane separation equipment.

Or

(b) Describe any two methods of chemical food processing.

15. (a) Discuss the features of any two food standards.

Or

(b) Examine the role of PFA and BIS.

Part C (3 × 10 = 30)

Answer any **three** questions out of **five**.

16. Explain the principles of food.

17. Discuss the food adulteration.

18. Describe the classification of drugs.

19. Write an essay on the physical principles of food processing.

20. How will you assess the quality of food materials?

AFF–1062

Sub. Code
4MBO3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

PLANT PHYSIOLOGY AND BIOPHYSICAL CHEMISTRY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Hydrophonics.

2. Abiotic stress.

3. Photorespiration.

4. Leghaemoglobin.

5. Ethylene.

6. Absorption spectrum.

7. Enthalpy.

8. Free energy.

9. X-ray diffraction.

10. Surface plasma resonance.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Give a detailed account of apoplast, symplast concept pertaining to water absorption.

Or

(b) Explain the mechanism of ascent of sap in plants.

12. (a) Give a detailed account of Calvin cycle.

Or

(b) Give an account of electron transport system in detail.

13. (a) Define vernalization. Discuss the various factors influencing vernalization.

Or

(b) Explain the causes of dormancy in plants and its significance.

14. (a) Write short notes on covalent bonds.

Or

(b) Explain optical isomerism.

15. (a) Describe the principle, operational mode of Circular dichroism.

Or

(b) Explain the principles of Electron paramagnetic resonance.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the role of mineral nutrient elements in plants.

17. Explain various steps of TCA cycle.

18. Write a comparative account of the physiological effects of auxins, gibberellins and cytokinins in plants.

19. Explain the law of thermodynamics and their application on the living system.

20. Describe the role of NMR in molecular structure determination.

AFF–1063

Sub. Code
4MBO3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

PLANT BIOTECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Classical biotechnology.

2. S1 nuclease.

3. Name the aminoacids synthesized by T-DNA.

4. *Vir* gene.

5. Callus induction.

6. Artificial seeds.

7. Cross protection.

8. Satellite RNA.

9. Trade Mark.

10. Biosafety.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the role of reverse transcriptase.

Or

(b) Give a brief account on the various types of vectors used in genetic engineering experiment.

12. (a) Describe the structure of Ti plasmid.

Or

(b) Explain the method of foreign DNA introduction into plant cell by micro and macro injection methods.

13. (a) Give a brief account of methods of sterilization.

Or

(b) Briefly describe the various approaches for micropropagation.

14. (a) What is biomethanogenesis? What its role in biogas production?

Or

(b) State your opinion about the patenting of GMOs.

15. (a) Explain in detail the basics of IPR.

Or

(b) Briefly explain the concept of containment in rDNA research.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write about the principle and application of PCR.

17. Describe about the strategies employed for genetically engineered plants for herbicide resistance.

18. Describe the mode of production of haploid plants and their various applications for crop improvement.

19. Describe the production of organic acids.

20. Discuss trade secret and copyright.

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AFF–1064

Sub. Code
4MBO3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

RESEARCH METHODOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Fixation.
2. Freeze fracture.
3. Autoradiography.
4. Nitrocellulose membrane.
5. Gel electrophoresis.
6. AFLP.
7. Applied Research.
8. Abstract.
9. Reprints.
10. Citation index.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Trace the pathway of light in a light microscope.

Or

(b) State the advantages and limitations of Electron microscopy.

12. (a) Explain the principle, procedure and application of Fluorescent in situ hybridization.

Or

(b) What are scintillation counter? Explain the types and their significance.

13. (a) Describe the procedure and applications of polyacrylamide gel electrophoresis.

Or

(b) Describe the procedure and applications of RAPD.

14. (a) Explain the need and functions of review of literature.

Or

(b) What is research design? What are its main features?

15. (a) Define the term editing and explain its various types.

Or

(b) Discuss in detail the various types of diagrams used in the research.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write an essay about the scanning electron microscope.

17. What are radio isotopes? Explain their use in tracer technology

18. Describe DNA sequencing methods.

19. What are the types of research? Explain them in brief.

20. Discuss the format of thesis.

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AFF–1065

Sub. Code
4MBO3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

Elective — BIORESOURCE TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define the following:

1. Photochemical smog
2. Biosphere
3. Solar cells
4. Tidal energy
5. Natural gas
6. GAIL
7. *Jatropha*
8. Gasifiers
9. Eutrophication
10. EIA.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Classify the natural resources.

Or

(b) Examine the indirect solar energy.

12. (a) Give an account of geothermal energy.

Or

(b) How will you account for the conversion of heat energy?

13. (a) Describe briefly the origin and development of coals.

Or

(b) Explain the composition of petroleum.

14. (a) Explain the safety measures to be adopted for the disposal of radioactive materials.

Or

(b) How will you account for the pyrolysis of biomass?

15. (a) Bring out the importance of bioremediation.

Or

(b) Examine the concept of biomagnification.

Part C $(3 \times 10 = 30)$

Answer any **three** questions out of five.

16. Write an essay on agroforestry.

17. Critically examine the features and application of wind energy.

18. Discuss the environmental problems associated with petroleum products.

19. Examine any four methods of radioactive waste disposal.

20. Analyse the advantages and benefits of renewable energy resources.

AFF–1066

Sub. Code
4MBO3E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

Elective — HORTICULTURE AND LANDSCAPPING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define the following:

1. Olericulture.
2. Drip irrigation.
3. Edges.
4. Terrace garden.
5. Hanging basket.
6. Bonsai plants.
7. Leafy vegetables.
8. Compost pit.
9. Pulsing.
10. Monoculture.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the scope of horticulture.

Or

(b) Differentiate cutting from layering.

12. (a) Give an account of lawns.

Or

(b) Describe the aesthetics of water garden.

13. (a) Compare training and pruning.

Or

(b) Examine the flower decoration.

14. (a) How you classify the vegetables?

Or

(b) Prepare the layout of a model kitchen garden.

15. (a) How will you induce flowering in a crop plant?

Or

(b) Explain the importance of teak plantation.

Part C $(3 \times 10 = 30)$

Answer any **three** questions out of Five.

16. Write an essay on the any two methods of grafting.

17. How will you design a model college garden?

18. Explain the cultivation practices and commercial aspects of rose.

19. Critically examine the cultivation practices of tomato and potato.

20. Analyse the cultivation practices and commercial aspects of mango.

AFF–1067

Sub. Code
4MBO3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

Elective — PLANT TISSUE CULTURE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Surface sterilization.

2. Sucrose.

3. Mericloning.

4. Totipotency.

5. Micropropagation.

6. Artificial seeds.

7. Androgenesis.

8. Ovary Culture.

9. Biotransformation.

10. Elicitors.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write about the basic layout of tissue culture laboratory.

Or

(b) What is the principle of the preparation of aseptic plants?

12. (a) Give an outline of the method of cell suspension culture.

Or

(b) What do you understand by the term Cytodifferentiation?

13. (a) Give an account of factors affecting micropropagation.

Or

(b) How do the requirement in the culture medium differ for root differentiation and shoot differentiation?

14. (a) Give an account of *in vitro* pollination.

Or

(b) How can protoplast fusion be achieved for raising somatic hybrids?

15. (a) What do you understand by plant cell immobilization and what are its advantages for secondary metabolite production?

Or

(b) Give a brief account of benefits from root culture.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the two most important culture media used for growing plant tissues.

17. Explain the different methods of single cell culture.

18. Give an account of plant growth regulators used in tissue culture.

19. Describe the technique of anther and pollen culture for getting haploid plants. Discuss its utility in plant breeding.

20. Describe with some examples where *in vitro* cell and tissue culture systems are exploited commercially for the production of secondary metabolites.

AFF–1068

Sub. Code
4MBO3E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

Elective – HERBAL BOTANY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Poisonous plants.

2. Folklore medicine.

3. Allergens.

4. Alkaloid.

5. Pharmacology.

6. *Datura.*

7. Sennoside.

8. CIMAP.

9. Isapgol.

10. Antimalarial drugs.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the unique features of unani medicine.

Or

(b) Write down the importance of traditional and folklore medicines.

12. (a) What are the scope of pharmacognosy?

Or

(b) Explain the various methods of drug evaluation.

13. (a) What are organoleptic tests? Explain with examples.

Or

(b) List out the therapeutical uses of *Adadhoda*, *Piper and Azadirachta*.

14. (a) Discuss the utilization of Bacopa and *Rauvolfia*

Or

(b) Explain the cultivation technology of *Cassia Senna*.

15. (a) Give a detailed account on the medicinal plant *Terminalia.*

Or

(b) Discuss the source and uses of glycoside drug studied by you.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the Siddha medicinal system with its principle, diagnostic methods and application.

17. Describe the standardization and quality control of herbal drugs.

18. Discuss the therapeutical and pharmaceutical uses of *Centella*, *Allium* and *Ocimum*.

19. Elucidate the cultivation and harresting technology of *Rauvolfia.*

20. Explain the morphology and useful parts, source and uses of lipid, resin and alkaloid drugs studied by you.